

Functional genetic report · longevity

Oxytocin and Social Behaviour

Sérgio F · Report date: 02-10-2026

Genes of the oxytocin/vasopressin system and polygenic scores of loneliness, sociability and autism. Only genes.
As a rule of Sérgio F, neurotransmission genes (here, receptors and neuropeptides) do not have favourable/unfavourable judgment; the judgment is only in scores with direct connection to health (solitude).

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Summary

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SNPs analyzed

6

in the chip

3

imputed ($DR2 \geq 0.8$)

2

SNP not found in the data

4

Polygenic scores

Oxytocin and vasopressin are hypothalamus neuropeptides that regulate bonding, trust, reading of emotions and response to social stress. Out of the brain, oxytocin maintains muscle stem cells (its decline with age contributes to sarcopenia in animal models), protects bone and has an anti-inflammatory effect. Chronic loneliness is one of the strongest mortality factors known (comparable to tobacco).

In summary: very favourable social polygenic profile. The loneliness score (658 variants) is at the 8th percentile and that of sociability at the 94th percentile. In the genes of the oxytocin system, Sérgio F has the genotype OXTR rs53576 GG (the most associated with empathy in small studies) and the rare genotype rs2254298 AA; these receptors are described without judgment.

Favourable

- **Loneliness: the 8th percentile** (PGS001091, 658 variants) and the 19th percentile in the simple score of 6 loci.
- **Sociability: the 94th percentile** (9 loci of GWAS Catalog).

Unfavourable

- No score with unfavourable judgment. The genes of oxytocin and vasopressin have no judgment (neurotransmission rule).
- **Not visible:** the microsatellite RS3 of AVPR1A and the SNPs of the gene OXT (rs2740210, rs4813625).

Interpretation: The genetic predisposition to loneliness is low and for sociability is high. As social attachment is one of the strongest longevity factors, this profile is a point in favor, as long as routine (work, training, travel) leaves room for regular relationships.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 6 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 0 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +).

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

The loneliness score (Tanigawa 2022) is the most complete; the sociability score and the simple loneliness score use few leading variants. The autism score (Grove 2019) is informative and goes without judgment.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Loneliness (658 variants)	PGS001091	559 of 658 (85%)	p8	Favourable
Loneliness (GWAS Catalog score, 6 loci)	C_SOLIDAO	6 of 6 (100%)	p19	Favourable
Sociability (GWAS Catalog score, 9 loci)	C_SOCIAB	9 of 9 (100%)	p94	Favourable
Autism spectrum disorder (35 087 variants)	PGS000327	33 520 of 35 087 (96%)	p65	Typical / no judgement

Oxytocin receptor (OXTR)

OXTR is the oxytocin receptor (brain, muscle, bone, heart, uterus). It is the most studied social gene, but the effects of isolated SNPs are small and many have not replicated in large samples.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
OXTR	rs53576	GG	chip	A	0	0.34	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). OXTR (3 intron): the G allele was associated with more empathy and search for social support in small studies; large meta-analyses do not confirm consistent effect. (moderate evidence)	Typical / no judgement
OXTR	rs2254298	AA	chip	A	2	0.11	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). OXTR: allele A was associated with differences in social sensitivity and volume of the tonsil. (weak evidence)	Typical / no judgement
OXTR	rs1042778	GG	chip	T	0	0.37	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). OXTR (3'UTR): associated in small studies with prosocial behavior and parental response. (weak evidence)	Typical / no judgement
OXTR	rs237887	GG	chip	A	0	0.55	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). OXTR: associated with empathy and social memory. (weak evidence)	Typical / no judgement
OXTR	rs2268498	TC	chip	T	1	0.54	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). OXTR promoter: expression of the receiver and recognition of emotions (weak evidence)	Typical / no judgement

Favourable

- No judgment (neurotransmission rule)

Unfavourable

- No judgment (neurotransmission rule)

Interpretation: rs53576 GG: the genotype most associated with empathy, trust and search for social support in small studies (large meta-analysis do not confirm effect). rs2254298 AA: rare genotype (~1% of the reference population), studied in social sensitivity and tonsil volume, with inconsistent results. No judgment, by rule.

Production and release of oxytocin (OXT, CD38)

CD38 controls the release of oxytocin into the brain; the OXT gene encodes oxytocin itself.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CD38	rs3796863	GT	imputation DR2=0.94	T	1	0.33	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). CD38 (release of oxytocin in the brain): C allele is associated with less CD38 expression and less plasma oxytocin. (moderate evidence)	Typical / no judgement
CD38	rs6449182	CC	imputation DR2=1.00	G	0	0.23	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). CD38 (1 intron): G allele increases the expression of CD38 in immune cells. (weak evidence)	Typical / no judgement
OXT	rs2740210		SNP not found in the data	C	—	—	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). Oxytocin gene: associated with stress response and social behavior. (weak evidence)	SNP not found in the data
OXT	rs4813625		SNP not found in the data	C	—	—	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). Oxytocin gene: associated with cortisol and response to social stress. (weak evidence)	SNP not found in the data

Favourable No judgment (neurotransmission rule)	Unfavourable No judgment (neurotransmission rule)
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Interpretation: CD38 rs3796863 heterozygous (1 allele associated with more plasma oxytocin) and rs6449182 CC (typical expression in immune cells). The SNPs of OXT are not in the data. No judgment, by rule.

Vasopressin (AVPR1A, AVPR1B)

Vasopressin (AVPR1A in the brain, AVPR1B in the pituitary) regulates bond, aggressiveness and stress axis.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
AVPR1A	rs11174811	CA	chip	A	1	0.12	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). Vasopressin 1A receptor (3'UTR): associated with male bond and social behavior; the most studied RS3 is not read by the chip. (weak evidence)	Typical / no judgement
AVPR1B	rs28632197	CC	imputation DR2=0.94	—	—	—	Receptor/neuropeptide: without favourable/unfavourable judgment (Sérgio F rule for neurotransmission). Vasopressin 1B receptor (hypophysis): stress axis (ACTH). (weak evidence)	Typical / no judgement

Favourable

- No judgment (neurotransmission rule)

Unfavourable

- No judgment (neurotransmission rule)

Interpretation: AVPR1A rs11174811 CA (1 allele infrequent) and AVPR1B rs28632197 CC. The most studied variant (RS3, a repetition) is not read by the chip. No judgment, by rule.

Social polygenic scores: loneliness, sociability and autism

Complete genome or GWAS Catalog scores for loneliness, sociability and autism. Loneliness is directly linked to mortality and therefore has judgment; autism is without judgment (neurodivergence).

Polygenic scores of this theme: Loneliness (658 variants) (PGS001091): the 8th percentile **Favourable**; Loneliness (GWAS Catalog score, 6 loci) (C_SOLIDAO): the 19th percentile **Favourable**; Sociability (GWAS Catalog score, 9 loci) (C_SOCIAB): the 94th percentile **Favourable**; Autism spectrum disorder (35 087 variants) (PGS000327): the 65th percentile **Typical / no judgement**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
Favourable - Solitude p8 (PGS001091) and p19 (simple score) - Sociability p94 Unfavourable - None								

Interpretation: Autism score (35 thousand variants) at the 65th percentile: typical, without judgment. The three social scores agree among themselves.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Intercellular Communication	—	Low genetic loneliness; high sociability	Social attachment reduces the inflammatory response to stress.
Chronic Inflammation / Inflammation	—	P8 loneliness (chronic loneliness increases inflammation and cortisol)	Favourable.
Neuroinflammation	—	Low loneliness (less risk of cognitive decline associated with isolation)	Favourable.
Exhaustion of Stem Cells / Sarcopenia	—	—	Oxytocin maintains the satellite cells of the muscle; without genetic marker with judgment.
Endocrine dysregulation	—	—	Stress axis (AVPR1B) without judgment.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
Regular social connection	Keeping weekly meetings with friends and family and group activities (group training, club, volunteering): genetic predisposition helps, but the effect on longevity comes from real contact. suggestion (not applied)	Loneliness p8; sociability p94
Touch and oxytocin	Physical contact, pets and massage rise to endogenous oxytocin. suggestion (not applied)	Oxytocin system
Group training	Add social component to exercise (team, training partner): sums up two factors of longevity. suggestion (not applied)	High Sociability
Intranasal oxytocin	Do not use without indication: the effects in healthy adults are small and inconsistent. suggestion (not applied)	OXTR without judgment
Coordination with the plan	Melatonin as in the plan (0.3–1 mg), omega-3 3.2–4 g/day and D3 10 000 IU. No telomerase activators. suggestion (not applied)	V2 plan rules

Limitations

- The effects of OXTR, CD38 and AVPR1A SNPs are small and many have not replicated in large studies.
- The microsatellite RS3 of AVPR1A and the SNPs of the gene OXT are not in the data.
- Social scores measure predisposition, not behavior; the environment weighs much more.
- The percentiles compare with the reference population (n=633).
- Only genes.

Methods and sources

- Genotypes: Raw DNA of the chip (GRCh38, strand +), analyzed only locally; Missing SNPs (statistical imputation, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs). Locally calculated frequencies in the reference population (n=633).
- Scores: PGS001091 (solitude, Tanigawa 2022), PGS000327 (autism, Grove 2019) and simple scores of GWAS Catalog (local copy) for loneliness (6 loci) and sociability (9 loci). Percentages vs. reference population (n=633) with exactly the same variants.
- Main references: Rodrigues 2009 (OXTR rs53576, PNAS); Bakermans-Kranenburg 2014 and Li 2015 (OXTR meta-analysis); Feldman 2012 (CD38 and oxytocin); Walum 2008 (AVPR1A RS3, PNAS); Elabd 2014 (oxytocin and muscle stem cells, Nat Common); Holt-Lunstad 2010 (solitude and mortality, PLoS Med); Day 2018 (Loneliness Genetics, Nat Commun).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).